Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Home Assignment 1

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Question

Exercise 1 - Leapfrog
Study the leapfrog scheme for the advection equation.
a) Give the leapfrog scheme (centered scheme with 2nd order accuracy in space and time).
b) Derive the stability criterion.
c) Discuss the computational mode and how it can be avoided.
Exercise 2 - Upwind
Study the upwind scheme for the advection equation.
a) Give the upwind scheme (uncentered scheme with 1st order accuracy in space and time).
b) Derive the stability condition.
HINT: The upwind scheme consists of a forward and backward part!
Exercise 3 - Euler forward
Study the Euler forward scheme for the diffusion equation.
a) Derive the stability criterion of the Euler forward scheme for the diffusion equation:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = A\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2}.$$

b) Discuss how the time step should be chosen. Why is it good to choose a smaller time step than that given by the stability criterion? Hint: Study how the amplification factor depends on the wavelength, especially the highest frequencies.
c) Are the oscillations that may appear in this scheme a numerical mode?

1 First exercise

(a) The leapfrog scheme is defined as

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + c\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0, \tag{1}$$

where $x = j\Delta x$, $t = n\Delta t$ and $\mu = c\Delta t/\Delta x$ is the Courant number.

(b) Solving Equation 1 one obtains

$$u_j^{n+1} = u_j^{n-1} - \mu\left(u_{j+1}^n - u_{j-1}^n\right). \tag{2}$$

This can be solved by doing a Fourier ansatz (we assume a wave-solution). This gives us:

$$\begin{aligned} u_j^n &= u_0 e^{ik(j\Delta x - nc_D\Delta t)} \\ u_j^{n+1} &= u_0 e^{ik(j\Delta x - (n+1)c_D\Delta t)} = u_j^n e^{-ikc_D\Delta t} = \lambda u_j^n. \end{aligned}$$

Above we defined $\lambda = e^{-ikc_D\Delta t}$, which tells us if the iterations increases which each step or decreases with each step. We can do the same for the spatial steps and we thus obtain:

$$\begin{aligned} u_j^{n+1} &= \lambda u_j^n \\ u_j^{n-1} &= \lambda^{-1} u_j^n \\ u_{j+1}^n &= e^{ik\Delta x} u_j^n \\ u_{j-1}^n &= e^{-ik\Delta x} u_j^n \end{aligned}$$

Using this result we can rewrite as Equation 1 as

$$0 = \frac{\lambda u_j^n - \lambda^{-1} u_j^n}{2\Delta t} + c\frac{e^{ik\Delta x}u_j^n - e^{-ik\Delta x}u_j^n}{2\Delta x} \implies \lambda = -i\mu\sin(k\Delta x) \pm \sqrt{-\left[\mu\sin(k\Delta x)\right]^2 + 1}. \tag{3}$$

Again this is stable if $|\lambda| \leq 1$. Thus we check the size of λ. This gives us two scenarios; either the square-root is imaginary or not.

Case 1: The square root is real (i.e. that $[\mu \sin(k\Delta x)]^2 < 1$). This implies that the size of λ is:

$$|\lambda|^2 = [\mu \sin(k\Delta x)]^2 - [\mu \sin(k\Delta x)]^2 + 1 = 1,$$

i.e. $\lambda \leq 1$ and the solution is always stable!

Case 2: The square root is imaginary. This implies that expression inside the square-root is less then one, and $[\mu \sin(k\Delta x)]^2 > 1$, furthermore this also implies that $\mu \sin(k\Delta x) \geq 1$. We thus have something exceeding our bound of 1 plus minus something else. We thus always obtain one solution outside of our bond.

Answer

Since we cannot determine the value of the $\sin(k\Delta x)$, we must assume that it obtains its largest value $\sin(k\Delta x) = 1$ somewhere. We thus end up with the condition that the solution is stable if $\mu \leq 1$.

(c) The solution gave two roots meaning if $\mu = 1$ we have $\lambda_{1,2} = \pm 1$. Here $lambda_1 = 1$ is the good physical moe meaning that each step is of the same size, and same sign. However, the $\lambda_2 = -1$ is the computational mode and will give an error to the solution.

There are differential methods to avoid this computational mode. One way is to occasional use an euler forward step, to avoid the build up from the computational mode. Another common way is to use some kind of filter. A famous filter is the Robert-Asselin filter. This filter works by first applying ones leapfrog step, and then smoothing the solution by using

$$u_f^n = u^n + \gamma \left(u_f^{n-1} - 2u_n + u_{n+1} \right). \tag{4}$$

Above one has to choose a suitable smoothing parameter, γ, to obtain nice smoothing without loosing to much information. That is the drawback of this filtering, that the filter will remove some information (often amplitude) fro the solution, however this will dampen the oscillations fro the computational mode.

2 Second exercise

(a) The upwind scheme uses the fact that we know which direction propagation comes from. This gives

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \text{ if } c > 0 \tag{5}$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c\frac{u_{j+1}^n - u_j^n}{\Delta x} = 0 \text{ if } c < 0. \tag{6}$$

I.e. if the wind (the information) travels to the right we must use information from the left, and if the wind travels to the left we must use information from the right.

(b) To derive the stability condition for the upwind scheme, we need to consider each case separately.

Case $c > 0$: Using the positive case we can write:

$$u_j^{n+1} = u_j^n - \mu \left(u_j^n - u_{j-1}^n \right). \tag{7}$$

again using the Fourier ansatz we can use

$$\begin{aligned} u_j^{n+1} &= \lambda u_j^n \\ u_j^{n-1} &= \lambda^{-1} u_j^n \\ u_{j+1}^n &= e^{ik\Delta x} u_j^n \\ u_{j-1}^n &= e^{-ik\Delta x} u_j^n \end{aligned}$$

and write

$$\lambda u_j^n = u_j^n - \mu\left(u_j^n - u_j^n e^{-ik\Delta x}\right) \implies \lambda = 1 - \mu\left(1 - e^{-ik\Delta x}\right) = 1 - \mu\left[1 - \cos(k\Delta x) + i\sin(k\Delta x)\right]. \quad (8)$$

Taking the norm of this to evaluate if $|\lambda| < 1$ we obtain:

$$|\lambda|^2 = \left[1 - \mu + \mu\cos(k\Delta x)\right]^2 + \left[\mu\sin(k\Delta x)\right]^2.$$

We observe that the stability depends on μ. For $\mu \in [0, 1]$, we have $|\lambda| \leq 1$, implying stability. For $\mu > 1$, the scheme becomes unstable.

Case $c < 0$: Using the negative case we can write:

$$u_j^{n+1} = u_j^n - \mu\left(u_{j+1}^n - u_j^n\right). \quad (9)$$

Rewriting this with the Fourier ansatz gives

$$\lambda u_j^n = u_j^n - \mu\left(u_j^n e^{ik\Delta x} - u_j^n\right) \implies \lambda = 1 - \mu\left(e^{ik\Delta x} - 1\right) = 1 - \mu\left[\cos(k\Delta x) - 1 + i\sin(k\Delta x)\right]. \quad (10)$$

Taking the norm of this gives:

$$|\lambda|^2 = \left[1 + \mu - \mu\cos(k\Delta x)\right]^2 + \left[\mu\sin(k\Delta x)\right]^2.$$

We observe that the stability depends on μ as before. For $|\mu| \leq 1$, we have $|\lambda| \leq 1$, implying stability. For $|\mu| > 1$, the scheme becomes unstable.

Answer

We observe that the stability condition depends on μ. For both $c > 0$ and $c < 0$, the scheme is stable if $|\mu| \leq 1$. Thus, independently of the direction of propagation, the upwind scheme is stable provided the Courant number satisfies this condition.

3 Third exercise

(a) From the question it is given theta the Euler forward scheme is

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = A\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2}. \quad (11)$$

Solving this equation gives

$$u_j^{n+1} = u_j^n + \frac{A\Delta t}{(\Delta x)^2}\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right).$$

To tackle this problem we again use the assumption that the solutions is on the form

$$\begin{aligned} u_j^{n+1} &= \lambda u_j^n \\ u_j^{n-1} &= \lambda^{-1} u_j^n \\ u_{j+1}^n &= e^{ik\Delta x} u_j^n \\ u_{j-1}^n &= e^{-ik\Delta x} u_j^n. \end{aligned}$$

This transforms Equation 11 into

$$\lambda = 1 + \frac{A\Delta t}{(\Delta x)^2}\left(e^{ik\Delta x} - 2 + e^{-ik\Delta x}\right) = 1 + \nu\left[e^{ik\Delta x} - e^{-ik\Delta x}\right]^2 = 1 - 4\nu\sin^2(k\Delta x). \quad (12)$$

Above we defined the von Neumann constant $\nu = A\Delta t/(\Delta x)^2$. The worst case scenario for stability here is if $\sin(k\Delta x) = 1$, which gives

$$\lambda = 1 - 4\nu.$$

If we have that we only want $|\lambda| < 1$ we obtain

$$-1 \leq 1 - 4\nu \geq 1 \implies \nu \in \left[0, \frac{1}{2}\right].$$

Answer

The Euler forward scheme is stable if and only if $\nu \in \left[0, \frac{1}{2}\right]$, where $\nu = A\Delta t/(\Delta x)^2$.

(b) The stability criterion for the Euler forward scheme was proven to be

$$\nu = \frac{A\Delta t}{(\Delta x)^2} \leq \frac{1}{2}.$$

Solving for the time step one obtains an upper bound at

$$\Delta t \leq \frac{(\Delta x)^2}{2A}.$$

Choosing the largest time step ($\nu = 1/2$)gives the amplification factor

$$\lambda = 1 + [\cos(k\Delta x) - 1].$$

Since the shortest possible wavelength is $\lambda_{\text{wavelenght}} = 2\Delta x$, since we need at least to positions for a wave. We thus obtain the fastest frequency to be $k\Delta x \approx \pi$. This implies that for the highest frequencies $\lambda = -1$.

However, Equation 11 is suppose to describe a diffusion, but when $\lambda = -1$ we get no damping. Instead we only obtain a flipping sign change with this amplitude factor. Thus it can be smart to choose a smaller time step to obtain a physical damping of the system.

(c) These oscillations that appear when $\lambda = -1$ is NOT a computational mode. This is because the Euler forward scheme only uses two time steps n and $n + 1$, and not three like leapfrog does. Thus one does not obtain a quadratic equation with two roots, but a simple equation with one solution for λ. This is thus still part of the physical mode, but just behaves badly by oscillations in the short-wave limit.